SouthEast University

PROJECT PROPOSAL

Title:Restaurant Management System

Course Code: CSE282.6
Course Title: Introduction to Programming Language II (Java) Lab

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Project Title: Restaurant Management System

1. **Introduction**
   A Restaurant Management System is a software application designed to automate and manage the daily operations of a restaurant. This system will help in managing orders, customers, billing, and reports efficiently. The goal is to reduce manual work, minimize errors, and improve service quality.

2. **Objectives**
   - To develop a user-friendly system for managing restaurant operations
   - To implement all four OOP principles (Encapsulation, Abstraction, Inheritance, Polymorphism)
   - To manage customer orders and billing efficiently
   - To generate reports such as daily sales and order summaries
   - To handle errors using structured exception handling (try-catch)

3. **System Features**
   - User Registration/Login System
   - Menu Management (Add, Update, Delete items)
   - Order Processing System
   - Billing System (Auto calculation)
   - Report Generation (Daily/Monthly)
   - Error Handling using try-catch

4. **OOP Concepts Implementation**
   - Encapsulation:
     Private data members with public getters and setters (e.g., Customer, Order class)
   - Abstraction:
     Abstract classes like User or Person to define common structure
   - Inheritance:
     Classes like Admin and Staff will inherit from User
   - Polymorphism:
     Method overriding

5. **Technology Stack**
   - Programming Language: Java
   - GUI: Java Swing
   - IDE: NetBeans / IntelliJ IDEA

- Database (Optional): File handling or MySQL

6. System Architecture

The system will follow a 3-layer architecture:

- Presentation Layer: Java Swing GUI
- Business Logic Layer: Service classes handling logic
- Data Layer: File/Database for storing records 8. Exception Handling Use of try-catch blocks for runtime errors Custom exceptions (e.g., InvalidInputException, OrderNotFoundException) Proper error messages displayed in GUI.

7. Expected Output

- Fully functional Restaurant Management System
- Interactive GUI
- Accurate billing and reporting system
- Proper implementation of OOP principles.

8. Conclusion

This project will enhance our understanding of Java programming, OOP concepts, and GUI development. It will also give practical experience in building real-world applications.